

Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Cloud Computing [CSE_4063]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A mid-sized healthcare startup, *MediLink*, is developing an AI-based patient monitoring platform that (5)
collects real-time health data from wearable IoT devices. The company expects heavy data traffic
during national health campaigns and needs to ensure:

A)

- High scalability to handle sudden data surges,
- Strong data privacy and security to comply with healthcare regulations,
- Low infrastructure maintenance due to limited in-house IT expertise, and
- Occasional data sharing with government research institutions for collaborative projects.

As a Cloud Computing Consultant, recommend the most appropriate cloud deployment model (Public, Private, Community, or Hybrid) for *MediLink*.

Apply the characteristics of the chosen model to justify your recommendation in relation to the organization's requirements.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A growing e-learning company, *EduWave Solutions*, offers online coding and data science courses (5)
to thousands of learners across the world. Recently, the company has been facing challenges in:

- Managing increasing user demand during live sessions,
- Providing reliable access to course materials at any time and from any location,
- Reducing the costs of maintaining physical servers, and
- Ensuring that students can access the platform seamlessly even when new updates or features are deployed.

As a Cloud Computing Consultant, identify and explain any three key characteristics or features of Cloud Computing that *EduWave Solutions* can utilize to overcome the challenges mentioned in the scenario.

Apply each feature to the situation clearly to justify its relevance.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) A startup plans to deploy multiple operating systems on limited hardware resources. (5)
a) Apply the concept of virtualization to suggest an efficient setup.
A) b) Demonstrate how virtualization software manages multiple OS instances concurrently.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A research lab wants to leverage virtualization for parallel data simulations requiring GPU (5)
acceleration.
a) Explain how hardware-assisted virtualization and custom hardware improve simulation

performance.

b) Propose a hardware-software configuration that supports both high throughput and flexibility.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3) A Bank is shifting its online banking system to a cloud environment using Service-Oriented Architecture (SOA). (5)

- A)
- a) Identify the two services that can be made reusable in this banking system.
 - b) Analyze how loose coupling helps when updating one of the services.
 - c) Recommend a standard protocol that can enable communication among these services.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) A smart hospital system relies on a private UDDI registry to support Service-Oriented Architecture (SOA). Critical services like Patient Records, Emergency Alerts, and Medical Device Monitoring must be automatically discovered by hospital applications in real time. (5)

- a) Analyze how UDDI's core data structures enable accurate and fast discovery of these services during emergencies.
- b) Evaluate how Subscription and Replication APIs help maintain reliability and up-to-date service access across multiple hospital branches.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4) A research institute currently uses a Grid computing system for scientific simulations but is planning to migrate to a Cloud platform to support new AI-based healthcare applications that require real-time processing and elastic scaling. (5)

- A)
- a) Analyze why the Grid system may struggle to meet the new AI application requirements, considering features such as resource sharing, throughput, and scalability.
 - b) Recommend how Cloud computing features like on-demand self-service, virtualization, and rapid elasticity would improve performance and resource management after migration.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) A cloud platform automatically reallocates virtual machines when CPU usage exceeds 80%, ensuring continuous performance without manual intervention. Explain how automated policy-based SLA management is applied in this situation and discuss one advantage of using such automation. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) An organization regularly scans its cloud servers using automated tools to detect outdated software, misconfigurations, and open ports that could be exploited by attackers. Apply your understanding of vulnerability assessment to explain how this process helps the organization enhance and maintain the security of its cloud environment in this scenario. (5)

- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

B) An IT organization uses role-based access control (RBAC) in its cloud environment, where employees can access only the data and applications relevant to their job roles. (5)

Analyze how Role-Based Access Control (RBAC) enhances security in a cloud environment by restricting access based on user roles. In the given scenario, explain how RBAC reduces the risk of unauthorized access and supports secure access management.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-

CSE_4063:Cloud Computing

2026

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4063:Cloud Computing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)